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Maths

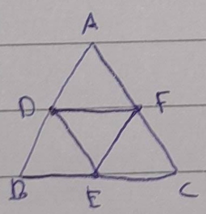
1) $\frac{AB}{DC} = \frac{AC}{DB}$ (As $\Delta ABC \sim \Delta DCB$)

Then, $AB \times DB = AC \times DC$

2) Distance $= \sqrt{(a-0)^2 + (b-0)^2}$
$= \sqrt{a^2 + b^2}$

3) $\angle BAC = \angle ADC$ (Given)
$\angle ACB = \angle DCA$ (common)
$\therefore \Delta ABC \sim \Delta DAC$ (By AA similarity)
$\Rightarrow \frac{CA}{CB} = \frac{CD}{CA}$

4) $P(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

5) 

$\Rightarrow DF \parallel BC$ and $DF = \frac{1}{2} BC$ (Mid-point theorem)
$\Rightarrow DF = BE$

In ΔABC and ΔEFD
$\angle ABC = \angle EFD$ (Opposite angles of llgm BDFE)
$\angle BCA = \angle EDF$ (Opposite angle of llgm DFCE)

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By AA criterion, $\Delta ABC \sim \Delta EFD$

If 2 triangles are similar, then ratio of their areas is equal to ratios of squares of corresponding sides.

$$\frac{\text{area}(\Delta DEF)}{\text{area}(\Delta ABC)} = \left(\frac{DF}{BC}\right)^2 = \left(\frac{DF}{2DF}\right)^2 = \frac{1}{4}$$

Hence, ratio of areas = ~~te~~ 1:4

6) $(0,y) = \left(\frac{6m_1 + (-3)m_2}{m_1 + m_2}, \frac{m_1 + 2m_2}{m_1 + m_2}\right)$

$\Rightarrow \frac{6m_1 - 3m_2}{m+n} = 0$ (As coordinate $x = 0$)

$\Rightarrow 6m_1 = 3m_2$

$\frac{m_1}{m_2} = \frac{3}{6} = \frac{1}{2}$

$\therefore$ Required ratio = 1:2

7) As $DE \parallel BC$, $\frac{AD}{DB} = \frac{AE}{EC}$

$\Rightarrow \frac{x+4}{x+3} = \frac{2x-1}{x+1}$

$\Rightarrow x+4(x+1) = 2x-1(x+3)$

$\Rightarrow x^2 + x + 4x + 4 = 2x^2 + 6x - x - 3$

$\Rightarrow x^2 + 5x + 4 = 2x^2 + 5x - 3$

$\Rightarrow -x^2 + 7 = 0$

$\Rightarrow x^2 = +7 \Rightarrow x = \sqrt{7}$

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8) Since, $(x, 0)$ is equidistant,

$\Rightarrow \sqrt{(-3-x)^2 + (4-0)^2} = \sqrt{(2-x)^2 + (5-0)^2}$

$\Rightarrow \sqrt{9 + x^2 + 6x + 16} = \sqrt{4 + x^2 - 4x + 25}$

$\Rightarrow \sqrt{x^2 + 6x + 25} = \sqrt{x^2 - 4x + 29}$

$\Rightarrow x^2 + 6x + 25 = x^2 - 4x + 29$ (By squaring both sides)

$\Rightarrow 10x = 4$

$\Rightarrow x = \frac{4}{10} = \frac{2}{5}$

$\therefore$ Coordinates are $\left(\frac{2}{5}, 0\right)$

9) $AD^2 = AB^2 + BD^2$

$\Rightarrow AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2$

$\Rightarrow AD^2 = AB^2 + \frac{1}{4} BC^2$

Also,

$CE^2 = BC^2 + BE^2$

$\Rightarrow CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2$

$\Rightarrow CE^2 = BC^2 + \frac{AB^2}{4}$

Then,

$AD^2 + CE^2 = AB^2 + \frac{BC^2}{4} + BC^2 + \frac{AB^2}{4}$

$\Rightarrow AD^2 + CE^2 = \frac{5}{4}(AB^2 + BC^2)$

$\Rightarrow AD^2 + CE^2 = \frac{5}{4} AC^2$ (In ΔABC, $AC^2 = AB^2 + BC^2$)

$\left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$

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$\Rightarrow \quad CE^2 = \frac{125}{4} - \frac{45}{4} = \frac{80}{4} = 20$

$CF = \sqrt{20}$ cm

10) X

SST

1) Status of women in India –

- Women are often confined to do only household work in many parts of India.
- Women are often not provided with equal educational and career opportunities.

2) Gandhiji used to say that religion can never be separated from politics. He believed that politics must be guided by ethics drawn from religion.

3) i) The constitution provides all individuals the freedom to profess, practice and propagate any religion.

ii) The constitution prohibits discrimination on grounds of religion.